

Background and Problem

Most treatment decisions rely on average effects from randomised controlled trials. Although RCTs remain the gold standard for causal inference, average effects from trial populations may not reflect individual treatment response. Subgroup analyses, while simple, have well-known limitations, including high false-positive and false-negative rates and limited value for individual decision-making ¹. Predictive approaches to treatment effect heterogeneity (PATH) are increasingly important in precision medicine, aiming to explain why patients respond differently to the same treatment by estimating individual outcomes with versus without treatment.

1. Implement and evaluate predictive HTE models across multiple clinical datasets

2. Compare risk-based and effects-based approaches in practice

3. Assess the clinical relevance and reporting implications of PATH analyses

Data Sources and Datasets

Data from our figures and further analyses is available from the following sources You can even use the bibliography the same way: Our data were taken from a cluster-randomised trial ², available from the [Irish Social Science Data Archive](#).

Early Results / Descriptive Statistics of Datasets

Usually you want to have a nice table displaying some important results that you have calculated. In `posterdown` this is as easy as using the `kable` table formatting you are probably use to as per typical R Markdown formatting.

You can reference tables like so: Table [1](#). Some basic summaries of the dataset are below:

Table 1: Table caption.

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
5.1	3.5	1.4	0.2
4.9	3.0	1.4	0.2
4.7	3.2	1.3	0.2
4.6	3.1	1.5	0.2
5.0	3.6	1.4	0.2
5.4	3.9	1.7	0.4
4.6	3.4	1.4	0.3
5.0	3.4	1.5	0.2
4.4	2.9	1.4	0.2
4.9	3.1	1.5	0.1

Figure [1](#), and Figure [2](#) below show the patterns in our dataset. Make sure that all the details in your plots will be legible when printed (legend text, axis text, and any labels)

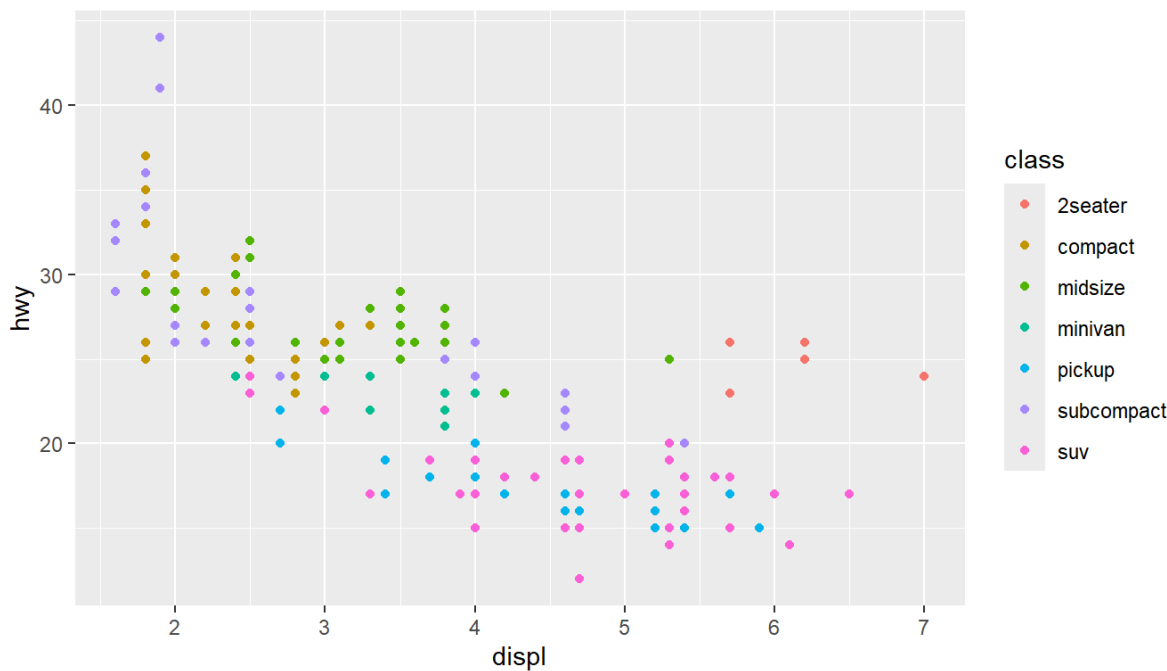


Figure 1: Great figure!

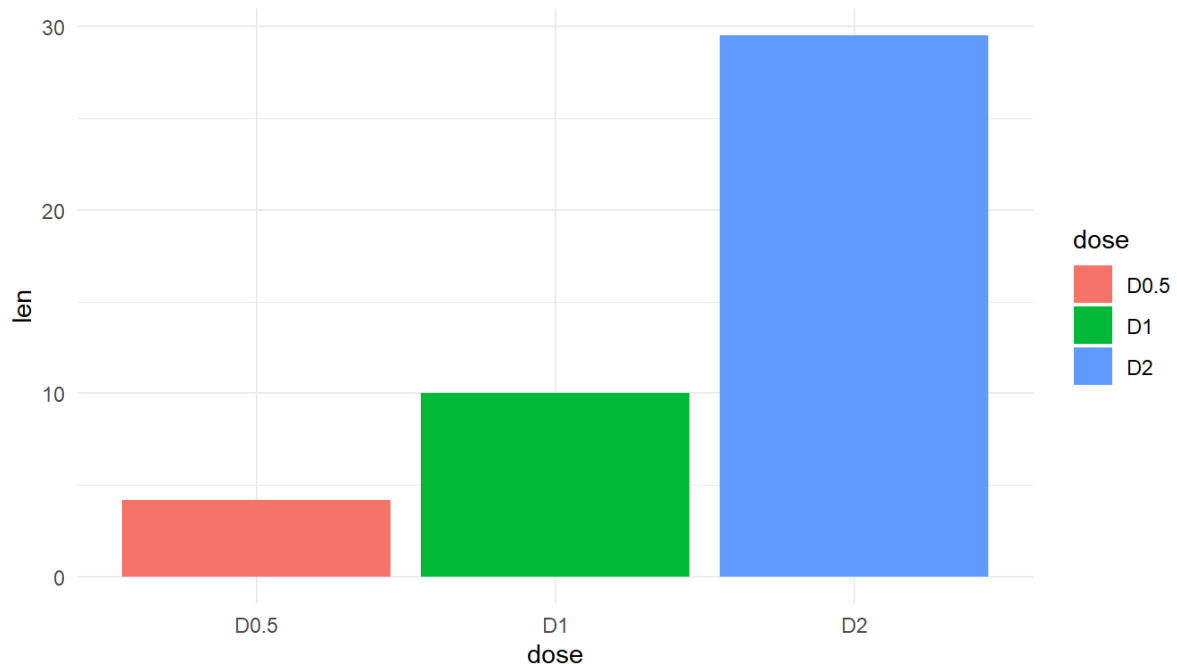


Figure 2: Amazing, right?!

You can even make your plots interactive for the HTML version of the poster. You can use the HTML poster for the presentation session, and the PDF poster will be printed - so be sure the static version looks okay.

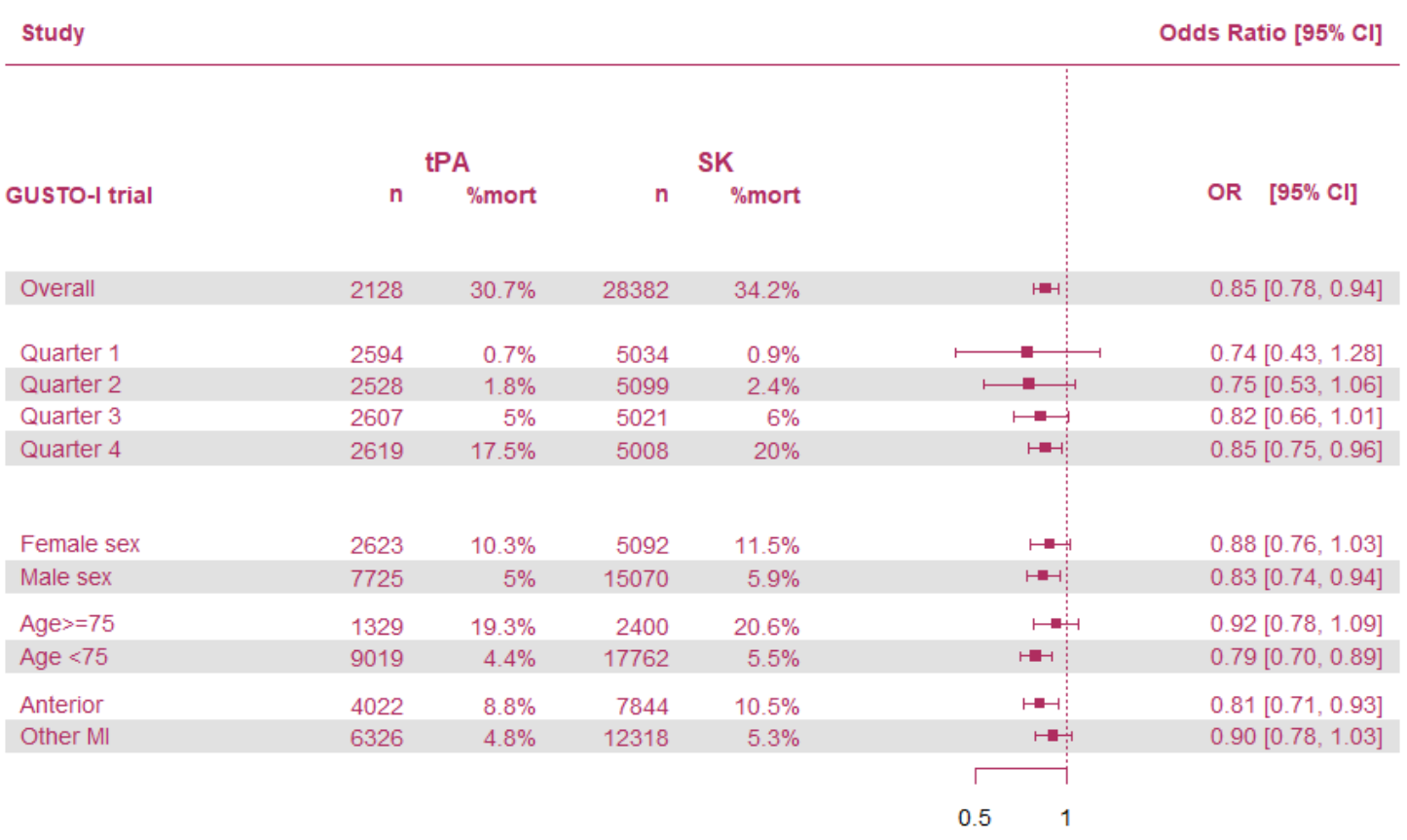


Figure 3: Figure 3: Forest plot of treatment heterogeneity

Next Project Steps

We plan to conduct further analysis using:

- Variable discombobulation ³

• Expand our minds with explosive machine learning ⁴.

We will use the `plasticanalysis` package for this.

GitHub

The code and datasets for this project can be viewed at our GitHub repository here: <https://github.com/>

References

1. Kent et al. 2020, doi: 10.7326/M18-3668 ## Objectives of Project↗

2. Murphy et al. 2005 doi: 10.1186/1468-6708-6-11↗

3. Massey et al. 2005 doi: 15.36.413↗

4. Smith et al. 1991 doi: 12.36.486↗